INCH-POUND

MIL-B-49430/3G(ER)
1 August 1991

SUPERSEDING
MIL-B-49430/3F(ER)
5 January 1990

# MILITARY SPECIFICATION SHEET

## BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5590/U

This specification is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49430(ER).

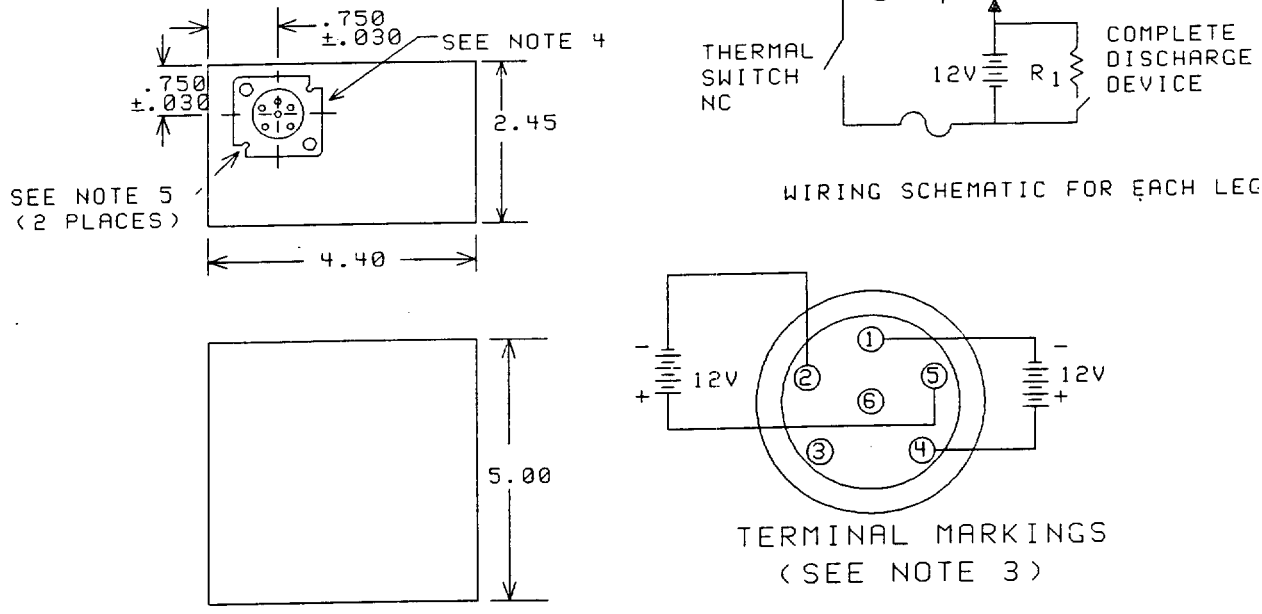



Figure 1. Battery dimensions and wiring schematic.

FSC 6135

AMSC N/A

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NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are -0.063, +0.000.

3. Terminal marking to be printed or embossed on jacket in suitable location near the socket on the terminal face of the battery.

4. The battery terminal shall consist of a miniature circular battery connector per Dwg. SC-C-179492 or equal. Connector shall be mounted in accordance with referenced drawing.

5. Connector socket body configuration shall accept two (2) each 0.156 inch diameter guidepost as shown. Depth of cavity shall be 0.657 inch minimum.

6. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 0.0625 inch below adjacent outside of the jacket when used initially, during, and after subsequent insertions of the mating plug during contractual testing.

7. $R_1$ value shall be between 140 ohms and 200 ohms and have a power rating of 2.0 watts.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 30.50 volts in 24 volt mode
15.25 volts in 12 volt mode

Minimum voltage for capacity: 20.00 volts in 24 volt mode
10.00 volts in 12 volt mode

The battery shall contain a 2.25 amperes non-replaceable slow blow fuse in each 12V leg wired in series with the cells in the negative leg.

A diode shall be wired in series with the cells in each individual 12 volt section to prevent charging current. See paragraph 3.5.5.2 of the basic specification.

Terminals: Six (6) hole socket type with no obstruction of any holes (see Fig. 1).

Weight (maximum): 2.33 pounds (1058 grams).



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Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

| Capacity Test* | Battery Capacity Requirements in Hours to 20.00 volts in Series 10.00 volts in Parallel |
|---|---|
| I & IP | 3.4 |
| L & LP | 2.2 |
| H & HP | 5.6 |
| HT & HTP | 5.4 |
| LT & LTP | 2.0 |
| T & TP | 3.0 |
| D & DP | 3.0 |

*NOTE: Sample batteries that are to be assigned to a capacity test shall be equally divided between the designated capacity test and the capacity test ending in P (e.g. I and IP). The odd battery, if present, is assigned to series discharge capacity test. For the capacity tests that do not have the P's, the battery's legs shall be discharged in series. For the capacity tests that have the P's, the battery's legs shall be discharged in parallel.

Initial voltage delay: During the I, H, HT, T, and D discharge tests of the two (2) 12 volt sections connected in series under the designated loads, initial voltage below the 20.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests initial voltage below 20.00 volts cannot exceed a 60 second duration. During the IP, HP, HTP, TP and DP discharge tests of the two (2) 12 volt sections connected in parallel under the designated loads, initial voltage below the 10.00 volt minimum voltage cannot exceed a 5.0 second duration. During LP and LTP discharge tests initial voltage below 10.00 volts cannot exceed a 60 second duration.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery on each leg as described in paragraphs 3.5.5 and 3.5.5.1 of the basic specification.

Complete discharge device: This device shall be inserted in the battery as shown on the wiring schematic and shall have the characteristics as described in paragraph 3.5.5.3 of the basic specification.



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Connector: The connector shall be in accordance with SC-C-179492 with the following exception: The durability test described in paragraph 4h shall be performed 100 times with SM-D-687888. Each of the metallic sockets in the connector (top view) shall conform to Figure 2. Each socket shall be cylindrical in shape and may have a single gap with a maximum opening of 30 degrees. The socket shall accept a number 16 pin. Compliance with this requirement shall be demonstrated as a part of incoming inspection.

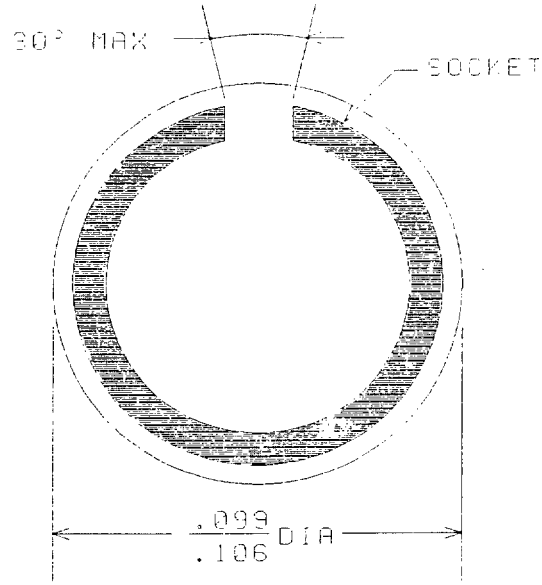



Figure 2. Socket (Top View).

METHOD OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage: Details on storage conditions for all specified capacity tests are described in basic specification (see 4.8.9).

(2) Discharge:

Cells: Discharge at 2.0 amperes constant current at 70±5°F for a minimum of 3.0 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 1.0 ampere constant current for 6.0 hours into voltage reversal.



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Batteries: For all capacity tests that do not end in P, except the H and HT, the batteries shall be discharged in the 24 volt mode by connecting pins 2 & 4 and discharged between pins 1 & 5 at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged in the 24V mode as described above, at two amperes for five minutes, followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts. For the I capacity test the batteries shall be force discharged at one ampere constant current for 6.4 hours immediately after zero volts is obtained.

For all capacity tests that end in P, except the HP and HTP, the batteries shall be discharged with the two 12 volt sections connected in parallel at four amperes constant current to zero volts. For the HP and HTP capacity tests the batteries shall be discharged with the two 12 volt sections in parallel at four amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts. For the IP capacity test the batteries shall be force discharged at two amperes constant current for 6.4 hours immediately after zero volts is obtained.

Closed-circuit voltage: Load resistance of 6.0 ohms shall be used between socket holes 1 and 4 and 2 and 5. Voltage shall be 11.70 volts or greater within twenty seconds.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
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(Project 6135-A234-03)



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